TODD CLARKSTON

Lead Software Architect | .NET / C# & Azure | Fintech, Healthcare & Enterprise Systems

Centennial, CO (Denver area) | 720-402-4347 | todd@toddclarkston.com

linkedin.com/in/toddclarkston | github.com/wireservers | Open to remote, hybrid, or on-site roles

PROFESSIONAL SUMMARY

Lead Software Architect and hands-on full-stack engineer with 20+ years delivering secure, enterprise-grade fintech, healthcare, and SaaS platforms with .NET, React Native, and Azure. Architected a greenfield EHR serving about 4,000 users and 2,700 patients across nine facilities while leading a five-person team, and built Federal Reserve bank-examination software used by 1,000 examiners nationwide. Applies AI in production, from Azure AI vision, document intelligence, and speech services to an LLM document-extraction platform with 99% accuracy processing about 100 documents in under a minute.

CORE COMPETENCIES

Software Architecture: Enterprise application architecture, solution design, full-stack development, REST APIs, microservices, SOA, domain modeling, SOLID principles, design patterns, API integration

Microsoft .NET: C#, .NET 10 / .NET Core, ASP.NET Core Web API, ASP.NET MVC, Entity Framework, LINQ, Dapper, ADO.NET, xUnit, NUnit, WPF, WCF, MAUI, Xamarin

Python: Flask, Django, REST APIs, ETL and data mapping, AWS Lambda, Azure Functions

Front End & Mobile: React Native, React Native Web, React, Next.js, Vue.js, Angular / AngularJS, TypeScript, JavaScript, NgRx, RxJS, Ionic, Xamarin, MAUI, responsive design, reusable component systems

Cloud & DevOps: Microsoft Azure, App Service, Azure Functions, Azure SQL, API Management, Entra ID, Bicep / ARM templates, Application Insights, Redis Cache, Azure DevOps, GitHub CI/CD, AWS, Docker, Kubernetes / AKS, Git

Security & Compliance: OAuth 2.0, OpenID Connect, SSO, MSAL, CIAM, OWASP Top 10, HIPAA, SOC 2-aligned and PCI DSS-aligned controls; PHI/PII handling

Healthcare Technology: Electronic Health Records (EHR/EMR), PACE, clinical workflows, appointment management, HL7 messaging, FHIR / Azure FHIR, healthcare interoperability

Financial Services: Banking and regulatory applications (Federal Reserve), bank-examination systems, account and transaction data, personal finance, PCI DSS-aligned controls, security and compliance approvals

AI & LLM Engineering: Azure Foundry Vision, Azure AI Document Intelligence, Azure Speech, Azure OpenAI, OpenAI API, vision OCR, document extraction, classification, summarization, multi-model fallback, Llama, Claude, prompt engineering, MCP, and agentic workflows

PROFESSIONAL EXPERIENCE

CenterLight Healthcare (via C2Q) — Lead Architect (Application Developer III)

Oct 2025–Present | New York, NY (Remote)

- Architected and lead development of a greenfield Electronic Health Record platform serving about 4,000 users and 2,700 patients across nine facilities, releasing to Development multiple times a day, UAT weekly, and Production each quarter.
- Lead a five-person engineering team, setting solution architecture, coding standards, code-review practices, technical direction, and delivery priorities while staying hands-on across the full stack.
- Integrated Azure Foundry Vision, Azure AI Document Intelligence, and Azure Speech into clinical and operational EHR workflows.
- Designed and developed a CMS-style form builder and workflow manager that replaced a confusing legacy form designer, cutting form creation from about an hour to 10 minutes; clinical and operational teams now build dynamic forms and multi-step workflows without code changes, and agents have embraced the new experience.
- Delivered patient and nursing appointment management, an admin portal, and an operational-status dashboard for clinical and support teams.
- Designed and implemented a custom CIAM/SSO server integrated with Microsoft Azure single sign-on and SOC 2-aligned security controls.
- Integrated HL7 messaging and Azure FHIR services for healthcare interoperability and HIPAA-compliant handling of protected health information.
- Built the cross-platform front end with React Native Web and the back end with .NET 10 Web API, C#, Entity Framework, and SQL Server; developed Azure workflows using Azure Functions and App Services.
- Managed full CI/CD pipelines in GitHub across Development, UAT, and Production environments, rolling back with deployment-slot swaps or by redeploying the previous build.
- Instrumented the code to detect Azure Key Vault, connection-string, and expired-key failures, so configuration problems surface in UAT logs before reaching Production.

- Use Claude Code with custom skills, MCP server integrations, and agentic workflows for code generation, reviews, automated testing, and development automation.

WireServers LLC — Owner & Principal Consultant

Jun 2009–2025 | Highlands Ranch, CO

Consulting practice, run alongside full-time roles, serving about 40 clients (10 repeat). Led a four-developer team and owned client relationships, solution architecture, full-stack delivery, cloud deployment, and ongoing application support.

Selected Client Engagements

- eLivelihood Dashboard: Oversaw development and maintenance of a care-management dashboard for organizing services for families. Served as DevOps Engineer, Full-Stack Engineer, and Technical Lead using Angular, Firebase, and Ionic.
- Swoovy: Co-developed and deployed the application as Full-Stack Engineer and Project Manager using Python, Django, React Native, and PostgreSQL.
- Series Code: Built Python services with Flask and Django over three years.
- Geolocation check-in app: Built an AWS Lambda geolocation service in Python for a geocaching-style app where hardware at each location site authenticates check-ins.
- Sent & Secure: Led development and maintenance of an end-to-end encrypted desktop application using SQL, REST APIs, Xamarin, WPF, Entity Framework, MAUI, and XAML.
- FitnessOne: Led front-end development in Angular with responsive UX, API integration, Material Design, and NgRx state management.
- Bionatal: Built API integrations between Ecwid, Klaviyo, and other platforms; used callbacks and Azure Service Bus to orchestrate marketing and sales-funnel workflows.
- BurstIQ: Integrated database-driven analytics into a React application; developed reusable components, hooks, themes, state management, and Nivo data visualizations.
- GuestsWanted.com: Large-scale vacation rental website for searching properties in any country, with extensive jQuery, mapping and location features, and a large property and user management system; featured on Wheel of Fortune in January 2011.
- CharityGolfNetwork.com: Delivered back-end database, middle-tier, and front-end development, wiring design mocks to MVC, jQuery, and mapping features.
- MLJ Document Manager (mljenvironmental.com): Built the client's document management application.

Architecture, Delivery & Leadership

- Integrated OpenAI into multiple applications for document OCR, classification, summarization, and structured extraction; designed multi-model architectures with provider fallback and token-usage tracking.
- Built Python ETL imports and data mapping, and Azure Functions in Python.
- Ran CI/CD and release processes on Azure DevOps and AWS (Lambda, API Gateway, CloudFormation, S3, EC2, Cognito, Route 53), deploying products to production multiple times a week, with Docker and Azure Kubernetes Service for containerized microservices.
- Mentored junior, mid-level, senior, and architect-level developers; led cross-team design reviews, code reviews, technical planning, and delivery coordination.

WellDyneRX (via TEKsystems) — Senior Software Developer

May 2018–Jul 2019 | Centennial, CO

Pharmacy Benefit Manager supporting high-capacity mail-order and retail pharmacy operations and processing about 15 million prescriptions annually.

- Developed full-stack pharmacy applications using MVC/C#, ASP.NET Web API, Knockout.js, jQuery, Bootstrap, SQL Server, LINQ, and Entity Framework.
- Automated multi-day prescription verification, order-fulfillment, and shipping workflows with RabbitMQ, replacing manual processing with queue-driven pub/sub orchestration.
- Developed HIPAA-compliant prescription-processing systems handling PHI, with HL7 messaging for pharmacy and healthcare data exchange, and remediated OWASP Top 10 and SonarQube-flagged security vulnerabilities.
- Helped redesign the architecture of a key enterprise application, increasing system performance and scalability by 30%.
- Reduced deployment time by 40% by automating Azure DevOps CI/CD pipelines and PowerShell deployments; led the migration of legacy systems to Docker containers, improving stability and simplifying releases.
- Provided architecture guidance and proof-of-concept development, and mentored junior and mid-level developers.

Federal Reserve Bank of Kansas City — Senior Software Engineer (Team Lead & Architect)

Jul 2014–Apr 2018 | Denver, CO

Delivered enterprise applications supporting Federal Reserve operations and regulated financial-services environments, including a mission-critical web application delivered on time and under budget.

Software Architect

- Coordinated across Federal Reserve districts and oversight departments to obtain security, quality, and usability approvals; led security-compliance integration across districts, reducing audit findings.
- Created coding standards and architectural guidelines adopted across multiple development teams, along with peer-review practices, technical documentation, and vendor-software evaluation processes.
- Brought onto the National team at the New York Fed to build software shared by all 12 Reserve Banks, starting with a common audit logger and an RBAC/ReBAC security model.
- With an architect counterpart, led the move from .NET Framework to .NET Core and built a national .NET Core component library.
- Applied secure-software practices and PCI DSS-aligned controls to financial applications and sensitive-data workflows.

Lead Developer

- Led a cross-functional delivery team of four developers, one software quality analyst, and one database administrator.
- Improved code-quality metrics by 35% by implementing SonarQube and peer-review practices.
- Increased team velocity by 25% through improved coordination and Agile best practices.
- Mentored 5+ junior developers, helping them advance in their careers, and supported knowledge transfer across teams.

Senior Software Engineer

- Designed SQL Server 2012 tables and stored procedures, improving data integrity, database maintainability, and application performance.
- Built the national Examiner Application, an MVC single-page app with RESTful services (C# 5.0, Angular 2, ASP.NET Web API 2, Entity Framework, TypeScript) giving 1,000 bank examiners across all 50 states access to each bank's information and history.

Pensco Trust — Web Developer III

Feb 2013–Dec 2013 | Denver, CO

Trust company specializing in self-directed alternative-asset investing. Client/Advisor Application: a single-page, RESTful application for managing investment accounts, with dashboards for asset allocation, account types, transaction history, activity, and alerts.

- Architected and launched a modular RESTful service layer that reduced integration time for new features by 35%.
- Refactored data access with Dapper, boosting query performance 40% across high-traffic endpoints.
- Improved front-end load times by 25% with lazy loading and optimized require.js module usage.
- As Scrum facilitator and Lead Developer, led the Agile transformation, improving sprint velocity and delivery predictability by 20%.
- Mentored developers who were later promoted to senior roles, and built a reusable UI component library with mustache.js and Bootstrap 3.
- Stack: MVC/C# 4.0, ASP.NET Web API, Ninject, Oracle 10g, Enterprise Library, Dapper, jQuery, OO JavaScript, mustache.js, pubsub.js, require.js, underscore.js, and Bootstrap 3.

Visa, Inc. — Software Engineer IV

Feb 2011–Aug 2012 | Highlands Ranch, CO

Global payments technology company. Consumer Website: a custom-branded prepaid-card site serving cardholders from over 1,000 financial institutions.

- Developed end-to-end solutions from the data access layer to the UI using ASP.NET/C# 4.0, WCF, SQL Server 2008, MVVM, and a custom security framework.
- Integrated a custom security framework that strengthened security and streamlined user authentication.
- Worked with the data team on two-way service communication and data integrity, reducing latency between services.

Quark, Inc. — Sr. Software Engineer

Dec 2008–Feb 2011 | Denver, CO

Publishing software leader known for QuarkXPress. Promote: a SaaS marketing platform for small and mid-sized businesses, with a WPF desktop client and public website.

- Reduced application load time by 30% through efficient database design and asynchronous WCF service calls.
- Led the migration of a legacy monolithic application to a modular SaaS architecture, improving maintainability and release cycles.
- Drove MVVM adoption across the engineering team, increasing UI test coverage and maintainability.
- Built WPF/MVVM clients, WCF and Windows Workflow Foundation services, and data layers with ASP.NET/C# 3.5 and the ASP.NET AJAX Toolkit; conducted code reviews and mentored junior developers.

eBags, Inc. — Sr. Software Engineer (Contract to Hire)

Jul 2007–Oct 2008 | Denver, CO

One of the largest online retailers of bags and luggage. Rewrote the consumer e-commerce website.

- Full-stack development with ASP.NET 2.0/C#, SQL Server 2005, user controls, and the ASP.NET AJAX Toolkit.

- Integrated Omniture, EdgeScape, Certona, Akamai, Scene7, and Endeca, and built payment integrations with PayPal, Google Checkout, and Bill Me Later.

SPIREMEDIA — Web Developer (Contract)
Nov 2006–Jun 2007

- Built a multimedia content management system with metadata tagging, news management, custom reporting, and user, role, and security administration, using ASP.NET/C# 2.0, SQL Server 2005, ASP.NET AJAX, NHibernate, and web services.

LANDesk Inc. — Sr. Web Developer (Contract to Hire)
Jul 2004–Aug 2006 | South Jordan, UT
Enterprise IT management, system configuration, and security software.

- Led a team that rebuilt the Windows-based server management application as a web application, becoming the company's web development team.
- Developed with n-tier architecture, ASP.NET 1.1, C# 1.1, SQL Server 2000, Oracle 9i, and web services.

J.P. Morgan Chase (via TEKsystems) — Sr. Software Engineer
Jan 2004–Jul 2004 | Salt Lake City, UT
The world's largest provider of commercial and consumer credit cards, with 25 Fortune 100 clients and over 100 million cardholders.

- Developed an online corporate credit card administration system (100+ functional specs) for managing employee cards, transactions, orders, and users, using n-tier ASP.NET 1.1, C# 1.1, and SQL Server 2000.

Certiport — Software Engineer (Contract)
Oct 2003–Jan 2004 | Salt Lake City, UT
Microsoft certification exam provider delivering exams in 128 countries and 20 languages.

- Built the online exam delivery system's UI and its data access, business façade, business rules, and configuration layers, and added Rijndael encryption to its enterprise security model (ASP.NET 1.1, C# 1.1, SQL Server 2000).

NEXTpage — Software Engineer (Contract)
Sep 2003–Oct 2003 | Salt Lake City, UT

- Delivered a knowledgebase administration site in four weeks for managing users, groups, configuration, and security, using ASP.NET/C# 1.1, XML, XSLT, server controls, and a data tier supporting SQL Server, Oracle 8i, FoxPro, and MS Access.

Teleperformance USA — Sr. Web Developer
Jun 2001–Sep 2003 | Salt Lake City, UT
One of the world's largest telecommunications companies and a leading international tele-services provider.

- Proposed and built a single intranet application that replaced multiple tools, including a call-floor scheduling tool and an IT ticket tracker.
- Streamlined the database structure, improving query performance and reducing load times by 30%.
- Developed dynamic content features that increased user engagement by 20%.
- Stack: ASP.NET, C#, VB.NET, XML, ADO.NET, DHTML, JavaScript, SQL Server 2000, and Oracle 8.1.7.

Programmers Market — Web Developer (Contract)
2001–2003 | Salt Lake City, UT

- Built an intranet financial tracker for sales, orders, inventory, spending, cost projections, and gains and losses, integrated with point-of-sale and ordering systems, using ASP 3.0, COM, JavaScript, and SQL Server 2000.

Alorica Inc. — Sr. Software Engineer
Jun 2000–Jun 2001 | Magna, UT

- Built call-tracking and correspondence-tracking systems that improved agent productivity and enabled per-item client billing and reporting.
- Improved data retrieval time by 30% with a streamlined database design, and reduced input errors by 25% with redesigned forms.
- Led the move from MS Access to SQL Server for better scalability and performance, using Visual Basic and SQL Server Desktop Engine.

Salar Corp. — Web Developer
Nov 1988–Jun 1999 | Denver, CO

- Managed technology across multiple retail stores, including the annual technology budget, point-of-sale and inventory systems, and databases.
- Launched online catalogs and e-commerce in 1996, contributing 10% of annual sales.
- Implemented software that increased operational effectiveness by 25% and reduced overhead, and trained more than 50 employees.

SELECTED PRODUCTS & OPEN-SOURCE ENGINEERING

React-Natives — Creator / Maintainer

Current | Open Source

GitHub: github.com/wireservers/react-natives | npm: @wireservers/react-natives

- Created and published a cross-platform library of 70+ reusable React Native and web components, used in production by BringTheBudget.
- Built the MIT-licensed library with React Native, NativeWind, Tailwind Variants, TypeScript, reusable theming, and developer-focused component APIs.

BringThe... Budget — Creator / Architect / Full-Stack Engineer

Current | bringthebudget.com

- Built and deployed an AI-powered personal-finance application that extracts structured accounts, transactions, and debts from Word, PDF, CSV, OFX, text, images, phone photos, and other document formats.
- Achieves 99% extraction accuracy and processes about 100 documents in under a minute, using Azure OpenAI vision, OCR, classification, summarization, multi-model fallback, and per-call token tracking.
- Owned end-to-end architecture and delivery using .NET 10, React Native Web, Entity Framework, SQL Server, Microsoft Azure, and an OpenAI-compatible Llama fallback.

BringThe... Diet & Food API — Creator / Architect

Current | bringthediet.com

- Built a healthy-recipes and diet-planning web app on Next.js 15 and React 19 with Microsoft Entra ID sign-in, deployed to Azure Web Apps through GitHub Actions.
- Designed the .NET 10 Food API on Azure App Service and Azure Cosmos DB (MongoDB API), with a repository pattern, paginated REST endpoints, Swagger, and cached counts that cut request-unit cost; deployed with Docker through Azure DevOps.

TECHNICAL SKILLS

C# / .NET: 20+ years — .NET Framework, .NET Core / .NET 10, ASP.NET Core, MVC, Web API, Entity Framework, LINQ, Dapper, ADO.NET

Python: 3+ years — Flask, Django, REST APIs, ETL imports and data mapping, AWS Lambda, Azure Functions, scripting

Web & Mobile Engineering: 10+ years — AngularJS through Angular 18, React, React Native, React Native Web, Next.js, Vue.js, TypeScript, JavaScript, NgRx, RxJS, Knockout.js, jQuery, Ionic, Xamarin, MAUI, HTML5, CSS3, SCSS

Databases: 20+ years — SQL Server, MySQL, PostgreSQL, Oracle, DynamoDB, MongoDB, Cosmos DB, Redis Cache, Firebase, MS Access, FoxPro; relational modeling, stored procedures, indexing, query optimization

Microsoft Azure: 10 years — App Service, Azure Functions, Azure SQL, API Management, VMs, Entra ID, SSO, custom CIAM, Key Vault, Azure FHIR, Service Bus, AKS, Application Insights, Redis Cache

DevOps & SDLC: 10 years — Azure DevOps, GitHub, Git, CI/CD pipelines, Bicep and ARM templates (infrastructure as code), automated builds, deployments, environment promotion, deployment slots and rollback, Docker, Kubernetes, PowerShell

Amazon Web Services: 3 years — Lambda, API Gateway, CloudFormation, S3, EC2, Cognito, Route 53, and cloud deployment patterns

Data Access & ORM: 14 years — LINQ, Entity Framework, Dapper, migrations, repository patterns, query optimization, and transaction management

Messaging & Integration: 8 years — RabbitMQ, Azure Service Bus, pub/sub messaging, event-driven workflows, REST/JSON, SOAP/XML, webhooks, callbacks, and third-party APIs

Healthcare Technology: 5 years — EHR/EMR, HIPAA PHI/PII handling, HL7, FHIR / Azure FHIR, prescription processing, clinical workflows, appointment management, PACE

Financial Services: Federal Reserve bank-examination and enterprise systems, financial applications, transaction and account data, personal finance, PCI DSS-aligned controls

AI / LLM Engineering: 3 years — Azure OpenAI, OpenAI API, GPT vision models, OCR, classification, summarization, structured extraction, Llama, Claude, Claude Code, MCP, prompt engineering, agentic workflows

Security: OAuth 2.0, OpenID Connect, JWT, MSAL, ASP.NET Identity, Entra ID, SSO, CIAM, RBAC / ReBAC, OWASP Top 10, SonarQube and Fortify scanning, SSL/TLS, Rijndael/AES encryption, secure SDLC, audit logging, HIPAA, SOC 2-aligned and PCI DSS-aligned controls

Architecture: Enterprise architecture, solution architecture, distributed systems, SOA, microservices, REST API design, design patterns, SOLID, object-oriented programming, domain modeling

Quality & Operations: xUnit and NUnit unit and integration testing, code reviews, automated tests, Application Insights observability, structured logging, troubleshooting, application support, performance tuning, documentation, runbooks

Earlier .NET & Web Platforms: ASP.NET Web Forms, ASP.NET AJAX, VB.NET, Visual Basic, ASP 3.0, COM, ADO.NET, WCF, Windows Workflow Foundation, WPF / XAML / MVVM, Ninject, Enterprise Library, NHibernate, XML / XSLT, mustache.js, require.js, underscore.js, pubsub.js

Integrations & Third-Party APIs: Stripe, PayPal, Google Checkout, Bill Me Later, Twilio, Mailchimp, Klaviyo, Ecwid, Boomi, QuickBooks, Omniture, Akamai, Scene7, Endeca, Certona, Firebase / AngularFire

UI/UX & Data Visualization: Responsive design, accessibility, wireframes, Material Design, Bootstrap, SCSS, D3.js, Chart.js, Nivo

Tools & Platforms: Visual Studio, VS Code, Git, Azure DevOps Boards, Postman, Insomnia, SonarQube, Fortify, Windows, macOS, Linux

Methodologies: Agile / Scrum (Certified ScrumMaster), Kanban, Extreme Programming (XP), Waterfall, full SDLC

EDUCATION & CERTIFICATIONS

- B.S., Computer Information Systems (Expected 2030), Metropolitan State University of Denver
- Microsoft Certified Professional (MCP)
- Scrum Master Certified — Scrum Alliance
- Recurring HIPAA privacy and security training